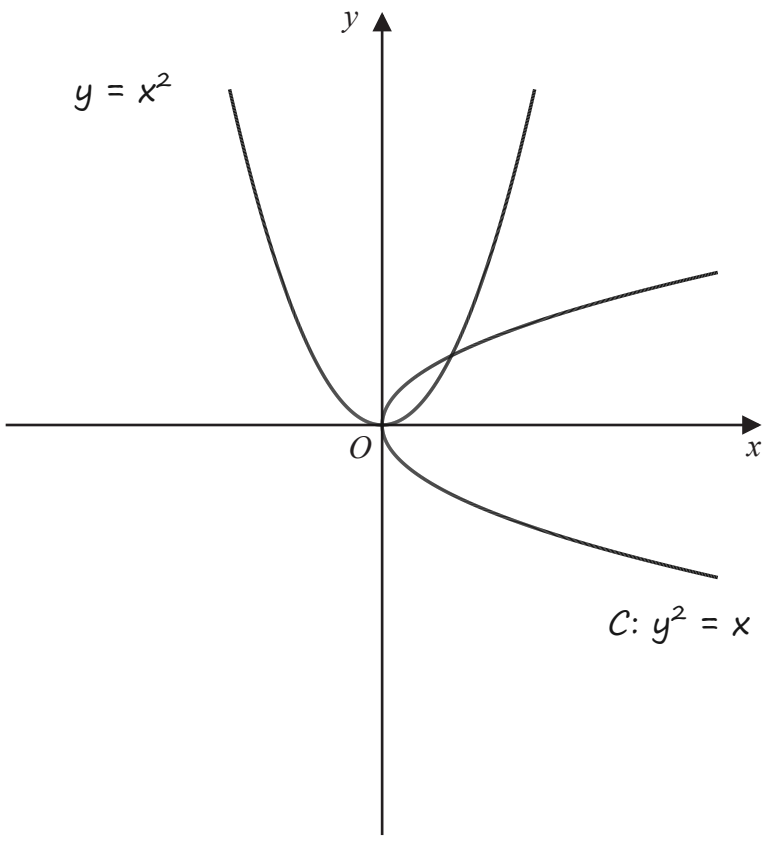
16 The graph of $y = x^2$ is reflected in the line with equation $y = x$ to give the curve **C**.

(a) Sketch the graph of $y = x^2$ and the curve **C**.
Clearly label the graphs.



(3)

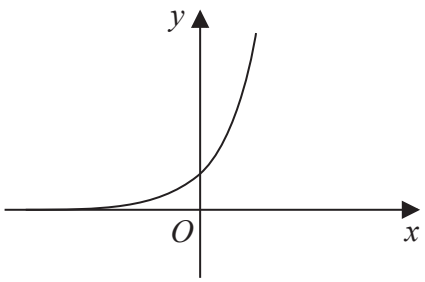
1 mark
Sketch $y = x^2$.

1 mark
Reflect in $y = x$.

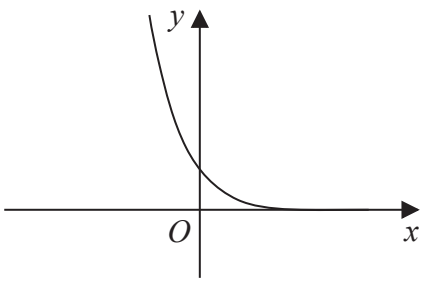
1 mark
Both pass through the origin.

Reflection swaps coordinates: $(u, u^2) \rightarrow (u^2, u)$

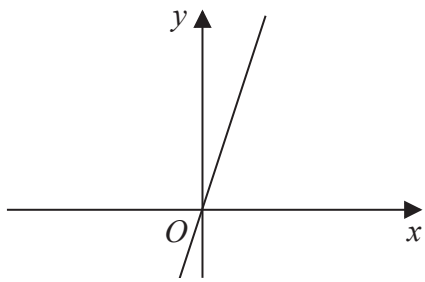
(b) Here are seven graphs.



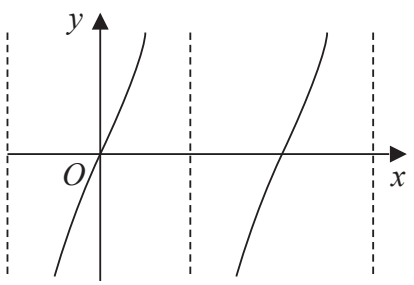
Graph A



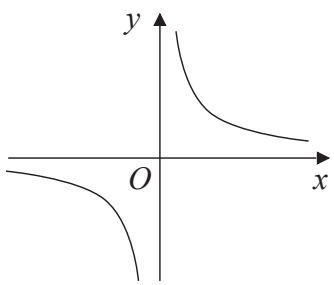
Graph B



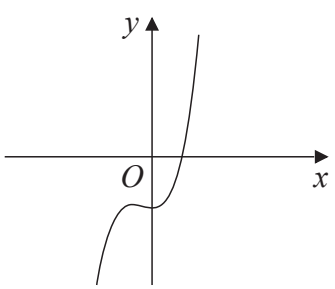
Graph C



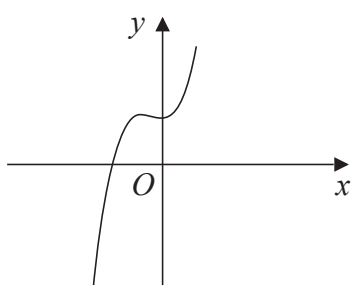
Graph D



Graph E



Graph F



Graph G

Complete the table below with the letter of the graph that could represent each given equation.

Equation	Graph
$y = 3^x$	A
$y = \tan x°$	D
$y = x^3 + x^2 + 2$	G
$y = \frac{3}{x}$	E

3 marks total
All four graphs matched.

(3)

(Total for Question 16 is 6 marks)